

CYCLEIQ DATA DICTIONARY & SCHEMA

REFERENCE

This technical asset rundown serves as the official data dictionary for the CycleIQ optimization model. It provides clear structural definitions, boundaries, and contextual constraints for fields loaded into the analytics environment from the core operational tables.

Important Temporal Mapping Key: As defined within the database layer constraints, the year dimension is structured as a binary integer flag where:

- **0** represents the **2021** financial year.
- **1** represents the **2022** financial year.

1. PRIMARY TRANSACTION TABLE SCHEMA

This table captures granular, hourly transactional snapshots of rider volumes cross-referenced with local environmental and calendar parameters.

FIELD NAME	DATA TYPE	DESCRIPTION / BUSINESS LOGIC CONSTRAINTS
dteday	Date	The specific date of execution matching the operational window (Formatted as DD/MM/YYYY).
season	Integer	Categorical indicator for meteorological seasons: 1 (Winter), 2 (Spring), 3 (Summer), 4 (Autumn).
yr	Integer (Flag)	Normalized year indicator variable. Map values strictly to: 0 = 2021, 1 = 2022.
mnth	Integer	Month index value ranging sequentially from 1 (January) through 12 (December).
hr	Integer	Hourly tracking dimension utilizing 24-hour block distribution (0 representing midnight to 23 representing 11 PM).
holiday	Boolean (0/1)	Binary switch stating whether the date corresponds to an officially recognized holiday (1 = True, 0 = False).
weekday	Integer	Day of the week index tracking cycle patterns (0 through 6).
workingday		

FIELD NAME	DATA TYPE	DESCRIPTION / BUSINESS LOGIC CONSTRAINTS
	Boolean (0/1)	Standard transactional flag. Evaluated as 1 if the day is neither a holiday nor a standard weekend (Monday–Friday corporate trading window), otherwise 0.
weathersit	Integer	Categorical rating of systemic weather severity (e.g., 1: Clear/Partly Cloudy, 2: Mist/Overcast, 3: Light Precipitation, 4: Heavy Storm/Severe Weather).
temp	Float	Normalized temperature reading scaled against region-specific maximum capacities.
atemp	Float	Normalized "feels-like" temperature metric accounting for localized wind chill and atmospheric factors.
hum	Float	Normalized humidity level percentile tracking moisture saturation points.
windspeed	Float	Normalized wind velocity readings mapped across a standardized continuous distribution.
rider_type	Varchar	Customer type tier distribution tracking baseline market composition (categorized as 'casual' or 'registered').
riders	Integer	The precise head-count volume of discrete customer checkouts recorded for that specified hour and demographic bucket.

2. FINANCIAL REFERENCE TABLE SCHEMA

This dimensional reference matrix isolates core financial parameters required to evaluate baseline profitability metrics.

FIELD NAME	DATA TYPE	DESCRIPTION / BUSINESS LOGIC CONSTRAINTS
yr	Integer (PK)	Primary Key linked directly to the main transaction year flag (0 = 2021, 1 = 2022).
price	Decimal (Currency)	The set base retail cost billed per ride instance for that specific year tier. Enforced in British Pounds (£3.99 for 2021, £4.99 for 2022).
COGS	Decimal (Currency)	Cost of Goods Sold. Represents the variable unit maintenance and operational overhead costs calculated per rental checkout (£1.24 for 2021, £1.56 for 2022).